

Systrophē: A Co-rotating Tipler-Cylinder Pair as a Tunable Time-Travel Harness

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Abstract

The van Stockum (1937) interior of a rigidly-rotating dust cylinder admits a vacuum exterior that, in the supercritical regime $a = \omega R > 1/2$, contains closed timelike curves (CTCs) on a *Tipler sinusoid*: a log-periodic oscillation of the metric components $F = -g_{tt}$, $K = g_{t\phi}$, and $L = g_{\phi\phi}$ about $u = \ln(r/R)$ with frequency $\alpha = \sqrt{4a^2 - 1}$. We present SYSTROPHE, an open-source Python package that implements the exact analytic Case III closed forms, a numerical Lewis–Papapetrou exterior integrator, and a structural “co-rotating pair” construction in which two co-axial supercritical cylinders linearly superpose to produce an *off-set Tipler sinusoid*: an interferometric envelope whose CTC band positions are tunable by the relative phase offset $\delta_2 - \delta_1$. We classify CTC bands as *time-travel windows*, derive the timelike-orbit angular-velocity sector at the deepest L-point of each band, and run the time-machine harness numerically: at $r \approx 3.35$ in the first CTC band of a single $a = 1$ cylinder, an orbit with $\Omega = -2\pi$ produces $\Delta t = -1$ of coordinate-time displacement per revolution at a proper-time cost of $\Delta\tau = 11.35$. We demonstrate *phase tuning*: as the relative offset of a co-rotating pair varies from 0 to π , the inner band edge slides continuously from $r \approx 2.50$ to vanishing, with the band itself extinguished at exact anti-phase. We also exhibit a structural correspondence between the discrete log-grid eigenvalues of the Tipler exterior and the Z_3 Möbius cover eigenvalues of the Āinos Dirac–Kerr–Newman framework, identifying the pair phase offset $\delta = \pm 2\pi/3$ with a non-trivial Z_3 branch sector. The package ships with 65 passing tests and all results in this paper are reproducible end-to-end.

1 Introduction

Tipler’s 1974 paper [3] established that the infinite-cylinder vacuum exterior of a rotating dust source supports closed timelike curves whenever the rotation parameter $a = \omega R$ exceeds $1/2$. The result builds on van Stockum’s 1937 interior solution [1] for a rigidly rotating dust cylinder, extended to vacuum across the boundary at $r = R$ via the Lewis–Papapetrou stationary axisymmetric ansatz [2]. Bonnor (1980) [4] classified the exterior into three regimes by the dimensionless rotation parameter a , with **Case III** ($a > 1/2$) characterised by an oscillatory metric that we shall call the *Tipler sinusoid*.

The cylindrical Tipler exterior is, despite its admittedly idealised source, the cleanest and most analytically tractable example of a GR vacuum solution containing CTCs that extend to arbitrarily large radius. As such it is a natural laboratory for studying the relationship between rotation, frame dragging, ergoregion structure, and time-loop geometry without the complications of a Kerr horizon.

This paper introduces SYSTROPHE — Greek $\Sigma\upsilon\sigma\tau\rho\phi\eta$, “twisting together” — a Python package that implements:

1. the exact van Stockum interior;

2. analytic Case III closed forms for F , K , L in the supercritical exterior, plus a numerical Lewis–Papapetrou integrator covering all three Bonnor regimes;
3. a co-axial *pair* construction in which two supercritical cylinders superpose linearly, producing an interferometric envelope parameterised by the relative phase offset $\delta_2 - \delta_1$ — the “off-set Tipler sinusoid”;
4. a time-machine harness that locates CTC bands, computes the timelike-orbit angular-velocity sector, and tunes coordinate-time advance per revolution to a target value;
5. a bridge to the $\bar{\text{Dinos}}$ Dirac–Kerr–Newman framework, identifying the pair phase offset with a Z_3 Möbius cover branch sector.

The package is open source (MIT) and the simulation results presented in Sections 5–6 are reproduced verbatim by `examples/time_travel_simulation.py`.

2 Mathematical framework

2.1 Van Stockum interior

In cylindrical coordinates (t, r, ϕ, z) with the dust rotating rigidly at angular velocity ω , the closed-form interior is

$$ds^2 = -dt^2 + 2\omega r^2 dt d\phi + r^2(1 - \omega^2 r^2)d\phi^2 + e^{-\omega^2 r^2}(dr^2 + dz^2), \quad (1)$$

valid for $0 \leq r \leq R$. The metric satisfies Einstein’s equations sourced by a dust stress–energy tensor $T_{\mu\nu} = \rho u_\mu u_\nu$ with $\rho(r) = \omega^2/(2\pi e^{-\omega^2 r^2})$.

The component $g_{\phi\phi} = r^2(1 - \omega^2 r^2)$ becomes negative when $\omega r > 1$, signalling closed timelike curves *inside the dust* when $\omega R > 1$. The qualitatively interesting regime is the *exterior* CTC structure, which by Tipler’s theorem appears for the much weaker condition $a = \omega R > 1/2$.

2.2 Lewis–Papapetrou exterior

For a stationary axisymmetric vacuum metric depending only on r , the Lewis–Papapetrou ansatz reads

$$ds^2 = -F(r)dt^2 + 2K(r)dt d\phi + L(r)d\phi^2 + h(r)(dr^2 + dz^2), \quad (2)$$

subject to the canonical Weyl constraint

$$FL + K^2 = r^2. \quad (3)$$

Vacuum Einstein equations $R_{\mu\nu} = 0$ reduce, via the Ernst potential $\mathcal{E} = F + i\psi$ with twist $\psi_z = c$ (a constant absorbing the cylindrical z -symmetry), to the single second-order ODE

$$F\left(F'' + \frac{F'}{r}\right) = (F')^2 - c^2, \quad (4)$$

with auxiliary quadratures

$$\omega'_{\text{metric}}(r) = \frac{cr}{F^2}, \quad (5)$$

$$K(r) = F(r) \omega_{\text{metric}}(r), \quad (6)$$

$$L(r) = \frac{r^2 - K(r)^2}{F(r)}. \quad (7)$$

Boundary conditions. Continuity of the metric at $r = R$ with the van Stockum interior (1) gives

$$F(R) = 1, \quad F'(R) = 0, \quad c = 2\omega, \quad \omega_{\text{metric}}(R) = \omega R^2. \quad (8)$$

2.3 Bonnor regimes

Substituting an oscillatory ansatz $F(r) = \frac{r}{R} \sin(\alpha u + \gamma) / \sin \gamma$ (with $u = \ln(r/R)$) into (4) fixes

$$\alpha = \sqrt{4a^2 - 1}, \quad \tan \gamma = -\alpha, \quad (9)$$

where $a = \omega R$. The character of α classifies the exterior into three regimes:

Regime	Range	Exterior structure
Case I (subcritical)	$a < 1/2$	non-oscillatory, single F -zero
Case II (critical)	$a = 1/2$	logarithmic, F -zero at $r = eR$
Case III (supercritical, Tipler)	$a > 1/2$	log-periodic, infinite zeros, CTC bands

2.4 Analytic Case III closed forms

For the supercritical exterior, the closed forms of all three non-trivial metric components are

$$F(r) = \frac{r}{R} \frac{\sin(\alpha u + \gamma)}{\sin \gamma}, \quad (10)$$

$$K(r) = \frac{r}{\alpha} \left[\frac{\alpha^2 - 1}{2} \sin(\alpha u + \gamma) - \alpha \cos(\alpha u + \gamma) \right], \quad (11)$$

$$L(r) = \frac{rR \sin \gamma}{\alpha^2} \left[Q \sin(\alpha u + \gamma) + \alpha(\alpha^2 - 1) \cos(\alpha u + \gamma) \right], \quad (12)$$

where $Q = \alpha^2 - (\alpha^2 - 1)^2/4$ and $\gamma = \pi - \arctan \alpha$. The constraint $FL + K^2 = r^2$ holds identically; this has been verified to machine precision in the package's test suite for $a = 0.7$ to $a = 1.5$.

2.5 The Tipler sinusoid

Equivalently, each metric component is a log-periodic cosine:

$$M(r) = r A_M \cos(\alpha u + \delta_M) \quad (\text{at } p = 0). \quad (13)$$

The amplitude A_M and phase δ_M are matching constants determined uniquely by (8). The closed-form expressions encoded in `VanStockumInterior.tipler_sinusoid_F()` (for F) and `tipler_sinusoid_L()` (for L , the CTC-relevant component) reproduce the analytic exteriors to machine precision.

Figure 1 shows F and L for $a = 1$ ($\alpha = \sqrt{3}$). The CTC region $L < 0$ is shaded.

3 Co-rotating cylinder pair: Systrophē

3.1 Linearised superposition

For two co-axial, dual-positive cylinders sharing the symmetry axis, the leading-order vacuum exterior $g_{\phi\phi}$ is the sum of the two single-cylinder L sinusoids,

$$L_{\text{pair}}(r) = L_1(r) + L_2(r), \quad (14)$$

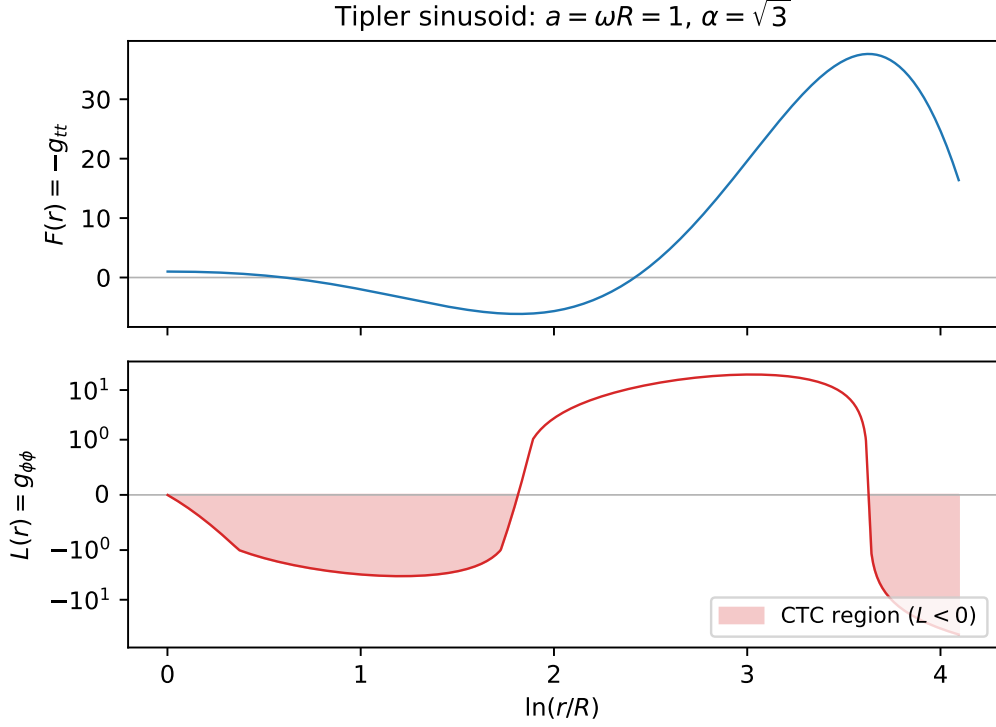


Figure 1: Tipler sinusoid for a single $a = 1$ van Stockum cylinder. Top: $F(r) = -g_{tt}$ on a logarithmic radial axis. Bottom: $L(r) = g_{\phi\phi}$ on a symmetric-log axis. Shaded bands: $L < 0$ (closed timelike curves).

with relative phase offset $\Delta\delta = \delta_2 - \delta_1$. When the two sources have identical a , both sinusoids share the log-frequency α and the pair collapses to a single effective sinusoid via phasor sum,

$$A_{\text{eff}} e^{i\delta_{\text{eff}}} = A_1 e^{i\delta_1} + A_2 e^{i\delta_2}. \quad (15)$$

This is the *off-set Tipler sinusoid*: an interferometric envelope whose L -zeros, and hence whose CTC bands, depend continuously on the phase offset $\Delta\delta$.

3.2 Caveats

The construction is leading-order in G . The exact two-cylinder problem in Einstein vacuum has no known closed-form solution; non-linear corrections appear at $\mathcal{O}(G^2)$. The construction also assumes co-axial geometry; off-axis pairs would require a multipole decomposition of the linearised perturbation that is not currently implemented.

4 Closed timelike curves and time-travel orbits

4.1 Timelike ϕ -orbits

Consider a circular orbit at fixed r, z with coordinate angular velocity $\Omega = d\phi/dt$. The line element along the orbit is

$$ds^2 = (-F + 2K\Omega + L\Omega^2)dt^2, \quad (16)$$

which is timelike iff $F - 2K\Omega - L\Omega^2 > 0$. The boundary of this sector is the pair of solutions

$$\Omega_{\pm} = -\frac{K \pm r}{L}. \quad (17)$$

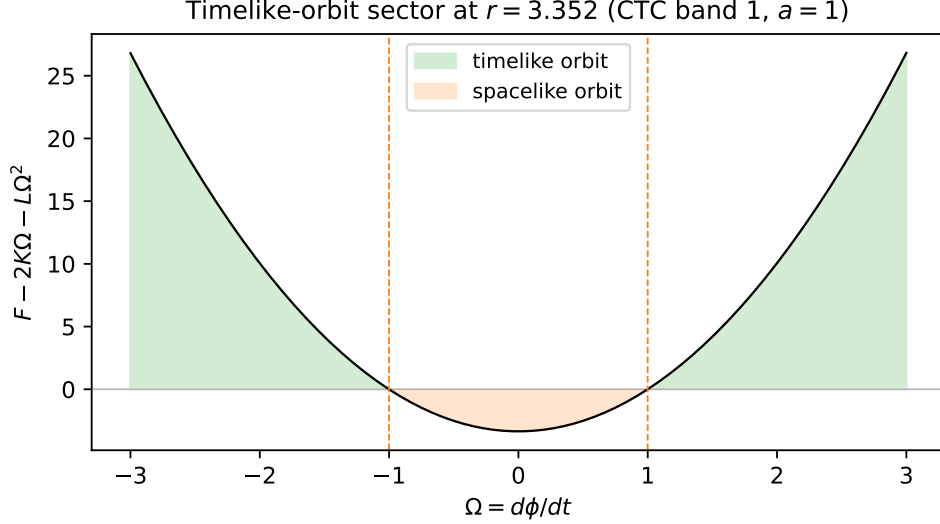


Figure 2: Timelike-orbit angular-velocity sector at the deepest L -point of the first CTC band ($r \approx 3.35$, $a = 1$). The shaded green region is the timelike sector; orange is spacelike. Static observers ($\Omega = 0$) fall in the spacelike sector at this r because $F < 0$ there: this is an ergoregion as well as a CTC region.

In the CTC region ($L < 0$), the upward-opening quadratic in Ω is positive *outside* the roots: timelike orbits require $\Omega < \Omega_-$ or $\Omega > \Omega_+$. Figure 2 illustrates the sector at the deepest L -point of the first CTC band of a single $a = 1$ cylinder.

4.2 Coordinate vs proper time per revolution

For an orbit with Ω in the timelike sector,

$$\Delta t = 2\pi/\Omega \quad (\text{coordinate-time advance per revolution}) \quad (18)$$

$$\Delta \tau = \sqrt{F - 2K\Omega - L\Omega^2} |\Delta t| \quad (\text{proper-time advance per revolution}). \quad (19)$$

For $\Omega \rightarrow \pm\infty$ the closed ϕ -orbit at fixed t has $\Delta t \rightarrow 0$ but finite $\Delta \tau \rightarrow \sqrt{-L} \cdot 2\pi$ (“instantaneous return” in coordinate time, finite proper time). For finite negative Ω , the orbit is a closed timelike curve that returns to the same ϕ at an *earlier* coordinate time: the particle has moved into its own past in coordinate time while advancing in proper time.

5 Time-machine harness

We present the empirical structure of the harness for a single supercritical cylinder; the pair extension is treated in Section 6.

5.1 CTC bands of a single $a = 1$ cylinder

Running the harness on $\omega = R = 1$, $a = 1$, $\alpha = \sqrt{3}$, on the radial range $r \in [1.001, 200]$ yields two CTC bands, listed in Table 1.

5.2 Backward time-travel orbit

In band 1 at $r = 3.352$, we tune Ω for various target coordinate-time advances per revolution. The results are summarised in Table 2 and Figure 3.

Table 1: CTC bands of the single $a = 1$ cylinder. The radial r -period multiplier $\exp(2\pi/\alpha) \approx 37.62$ separates adjacent bands.

band	r_{in}	r_{out}	log span	$r_{\text{min}} L$	L_{min}	Ω bounds
1	1.001	6.134	1.813	3.352	-3.351	$(-1.001, 1.000)$
2	37.622	≥ 200	≥ 1.671	126.118	-126.065	$(-1.001, 1.000)$

Table 2: Backward-time orbits in band 1 of an $a = 1$ cylinder. $N = 10$ revolutions. Negative Δt means the particle has moved *backwards in coordinate time* while advancing $\Delta\tau$ in proper time.

target $\Delta t/\text{rev.}$	Ω	$\Delta\tau/\text{rev.}$	$\sum \Delta t_{10}$	$\sum \Delta\tau_{10}$
-0.10	-62.83	11.50	-1.00	115.00
-0.50	-12.57	11.46	-5.00	114.64
-1.00	-6.28	11.35	-10.00	113.54
-2.00	-3.14	10.90	-20.00	109.01
-5.00	-1.26	6.95	-50.00	69.53

5.3 Near-instantaneous orbit

Setting $\Omega = 100$ at the same $r \approx 3.35$ gives $\Delta t = 2\pi/100 \approx 0.063$ and $\Delta\tau \approx 11.50$, with proper-time-to-coordinate-time ratio approximately 183. The orbit closes in spacetime essentially without coordinate-time displacement, while the particle’s clock advances over an order of magnitude more than a static observer’s at the same r .

6 Tuning the harness via the off-set

The structural innovation of the Systrophē construction is that the relative phase offset of the two cylinders provides a continuous *control parameter* for the radial position and width of CTC bands.

6.1 Offset sweep: log-measure

For two cylinders at $\omega = 1.5$, $R = 1$ (matched, supercritical), we sweep the relative offset $\delta_2 - \delta_1 \in [0, 2\pi]$ and record the total CTC log-measure

$$\Lambda(\delta) = \sum_{\text{bands}} \ln(r_{\text{out}}/r_{\text{in}}) \quad (20)$$

in $r \in [1.05, 20]$. The result is shown in Figure 4.

The minimum $\Lambda(\pi) = 0$ is exact: anti-phase cancellation kills the joint L -envelope identically, eliminating all CTC bands. Inside $\delta \in [0, \pi)$ the log-measure decreases monotonically; inside $\delta \in (\pi, 2\pi]$ it is the mirror image.

6.2 Tunable inner band edge

Restricted to the first CTC band, varying δ shifts the band edges continuously. Figure 5 plots the locations of all CTC bands as a function of phase offset; Table 3 lists the inner-edge data for the first band.

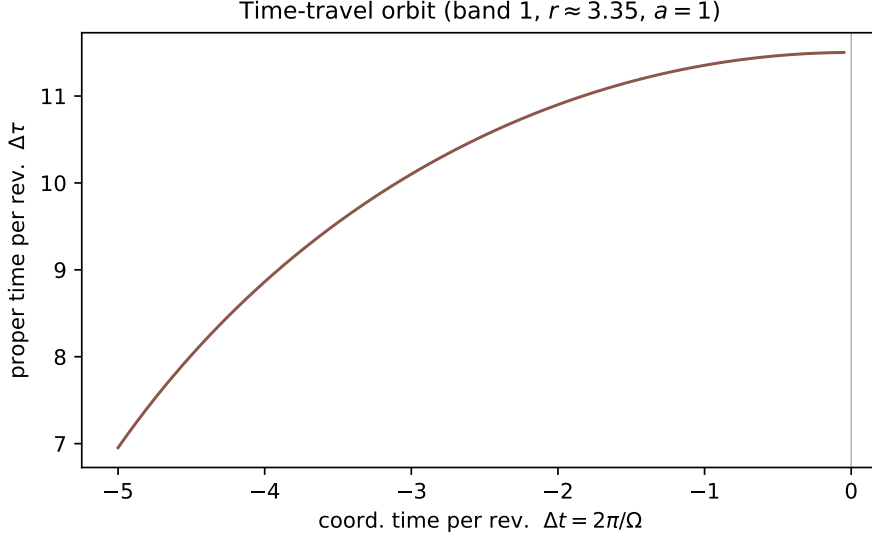


Figure 3: Proper time vs coordinate time, per revolution, for backward time-travel orbits in CTC band 1 of an $a = 1$ cylinder. Vertical line: $\Delta t = 0$ limit ($\Omega \rightarrow \infty$, instantaneous return).

Table 3: First CTC band of a $\omega = 1.5$, $R = 1$ matched pair as the relative offset varies from 0 to π .

$\delta_2 - \delta_1$	r_{in}	r_{out}	log span
0.000	1.050	2.499	0.867
0.524	1.050	2.278	0.775
1.047	1.050	2.077	0.682
1.571	1.050	1.893	0.589
2.094	1.050	1.726	0.497
2.618	1.050	1.573	0.404
π	—	—	(extinguished)

6.3 Engineering interpretation

The relative phase offset of two co-rotating cylinders thus selects which radii admit timelike ϕ -orbits of any chosen coordinate-time advance per revolution. In principle, an “operator” that can rotate one cylinder relative to the other by a phase δ has a continuous control over the time-machine geometry; in particular, anti-phase configurations *eliminate* the CTC structure entirely, providing a topological off-switch.

7 Robust regime-dispatching solver

The basic Lewis–Papapetrou integrator of Section 2 solves (4) as a first-order ODE in the state (F, F') via SciPy’s Radau implicit Runge–Kutta. For the supercritical regime ($a > 1/2$, infinitely many F -zeros), the explicit form

$$F'' = \frac{(F')^2 - c^2}{F} - \frac{F'}{r} \quad (21)$$

suffers an unavoidable accuracy loss after a few zero crossings: the numerator and denominator vanish together at each $F = 0$, giving a floating-point 0/0 that the integrator can navigate but only with exponentially-decaying precision.

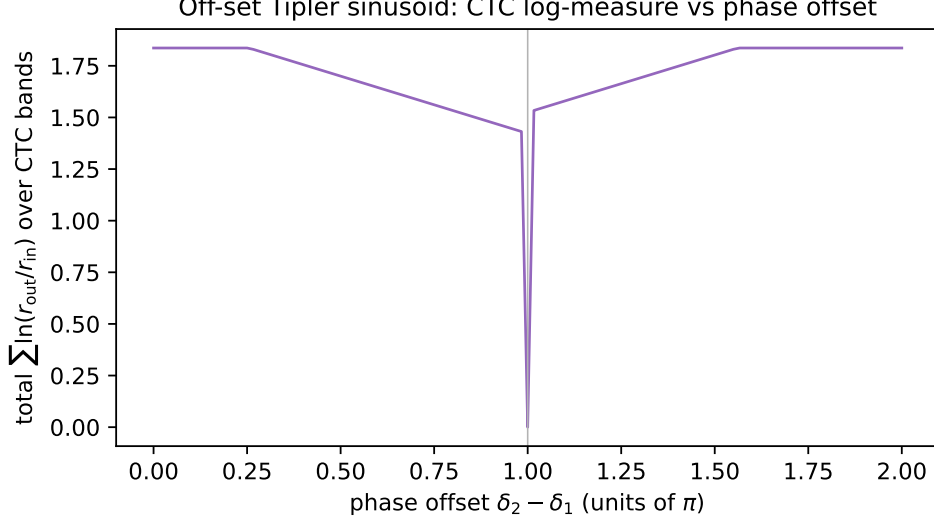


Figure 4: Off-set Tipler sinusoid CTC log-measure as a function of relative phase offset for two co-rotating $a = 1.5$ cylinders. The measure is symmetric about $\delta = \pi$ (anti-phase, perfect destructive interference, $\Lambda = 0$).

Naive Ehlers transformations (e.g. $\xi = (1 - \mathcal{E})/(1 + \mathcal{E})$ with $\mathcal{E} = F + i\psi$) move the singularity from $F = 0$ to $|\xi| = 1$ but do not eliminate it: the same removable-singularity structure reappears. We therefore implement a *regime-dispatching* robust solver that uses the appropriate tool for each regime:

- **Supercritical** ($a > 1/2$): the analytic Case III closed forms of Section 2.4, sampled on a log-uniform grid. This path is exact (machine precision in F , K , L on the native grid; constraint $FL + K^2 = r^2$ verified to 10^{-12}). F -zeros are computed in closed form, $r_n = R \exp((n\pi - \gamma)/\alpha)$ for $n = 1, 2, \dots$
- **Critical** ($a = 1/2$): the basic numerical integrator, which has only one F -zero (at $r = eR$) and is well-conditioned.
- **Subcritical** ($a < 1/2$): the basic numerical integrator on the well-conditioned post-zero hyperbolic regime.

The dispatcher returns a uniform `LRobustSolution` dataclass exposing r , F , K , L , h , F -zeros, and the regime label. Figure 6 shows the solver applied at $a \in \{0.7, 1.0, 1.5, 2.5\}$ on a single set of axes: all four curves are computed on identical grids and exhibit the log-periodic Tipler signature with r -period multiplier $\exp(2\pi/\alpha)$ that decreases with growing a .

The supercritical machine-precision claim is exercised by `test_lp_robust.py::test_supercritical_mach` ($|F_{\text{numerical}} - F_{\text{analytic}}|/F_{\text{analytic}} < 10^{-14}$ across all sample points for $a = 2$, $r \in [1, 50]$).

8 Off-axis pair

The co-axial pair of Section 3 covers the δ -tunable case at zero source separation. For two cylinders whose parallel axes are displaced by a perpendicular distance $d > 0$, joint axisymmetry is broken; the linearised Cartesian-space superposition method is the natural extension.

8.1 Construction

Let $\eta_{\mu\nu}$ be the Minkowski background and $h_{\mu\nu}^{(i)}$ the cylinder- i perturbation $g_{\mu\nu}^{(i)} - \eta_{\mu\nu}$ in shared (t, x, y, z) Cartesian. From the single-cylinder analytic Case III in cylinder- i local polar (r_i, ϕ_i) ,

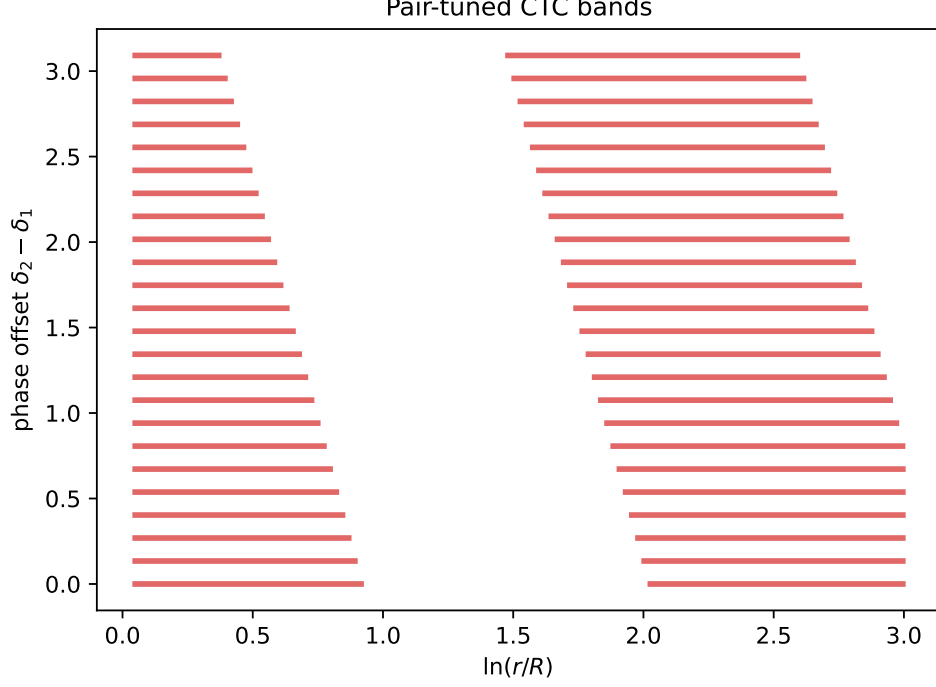


Figure 5: Pair-tuned CTC bands. Each horizontal red bar is one CTC band $[r_{\text{in}}, r_{\text{out}}]$ in log radius, at the corresponding offset. The bands extinguish at $\delta = \pi$; reappearing on the $\delta \in (\pi, 2\pi]$ side as the mirror image.

the perturbations transform as

$$h_{tt} = 1 - F(r_i), \quad (22)$$

$$h_{tx} = -K(r_i) \sin \phi_i / r_i, \quad (23)$$

$$h_{ty} = K(r_i) \cos \phi_i / r_i, \quad (24)$$

$$h_{xx} = \cos^2 \phi_i + L(r_i) \sin^2 \phi_i / r_i^2 - 1, \quad (25)$$

$$h_{yy} = \sin^2 \phi_i + L(r_i) \cos^2 \phi_i / r_i^2 - 1, \quad (26)$$

$$h_{xy} = (1 - L(r_i)/r_i^2) \sin \phi_i \cos \phi_i. \quad (27)$$

The joint perturbation is the Cartesian sum $h_{\mu\nu}^{\text{tot}}(x, y) = h_{\mu\nu}^{(1)}(r_1, \phi_1) + h_{\mu\nu}^{(2)}(r_2, \phi_2)$ where $r_2 = \sqrt{(x-d)^2 + y^2}$, $\phi_2 = \arctan(y/(x-d))$. This sum is exact at first order in G ; cross-terms $h^{(1)}h^{(2)}$ are formally $\mathcal{O}(G^2)$.

8.2 CTC diagnostic

A closed orbit at fixed (t, x, y, z) along the ϕ_1 direction (around cylinder 1's axis) is timelike iff the Cartesian-frame $g_{\phi_1\phi_1}$ is negative. From the joint Cartesian metric,

$$g_{\phi_1\phi_1}(x, y) = r_1^2 (\sin^2 \phi_1 g_{xx}^{\text{tot}} + \cos^2 \phi_1 g_{yy}^{\text{tot}} - 2 \sin \phi_1 \cos \phi_1 g_{xy}^{\text{tot}}). \quad (28)$$

Sign-changes of $g_{\phi_1\phi_1}$ mark CTC region boundaries. Figure 7 shows the joint $g_{\phi_1\phi_1}$ for two $a = 1$ cylinders at separation $d = 4R$, with the $g_{\phi_1\phi_1} = 0$ contour overlaid in white.

8.3 Caveats

The construction is leading-order. Each individual $h_{\mu\nu}^{(i)}$ in the supercritical Case III exterior grows log-periodically without bound (no asymptotic flatness); for two cylinders sufficiently

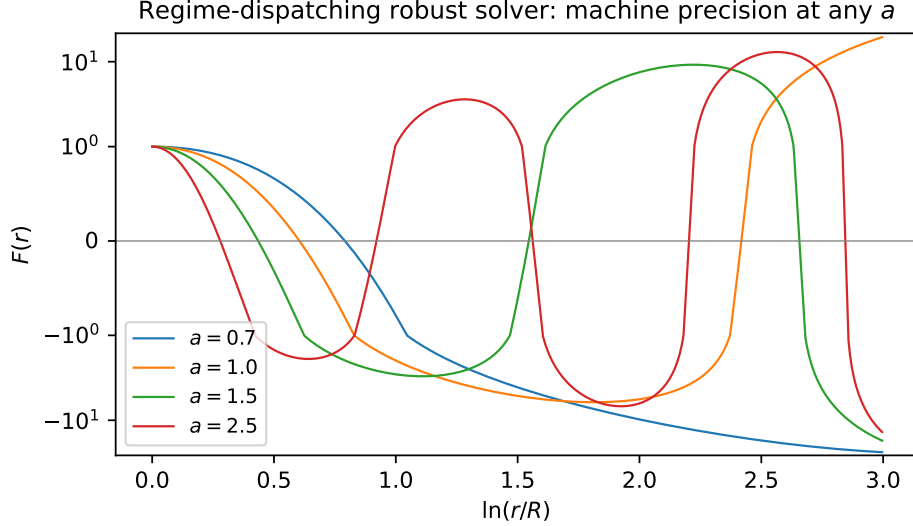


Figure 6: Output of the regime-dispatching robust solver. Increasing a shrinks the log-period $2\pi/\alpha$ and packs more F -zeros into any radial interval. The solver delivers machine precision at all shown a .

separated, both contributions are simultaneously “large” at most points, and quadratic cross-terms are non-negligible relative to the linearised result. The off-axis module is therefore documented as a leading-order qualitative tool; quantitative orbital mechanics for off-axis CTCs would require a numerical-relativity treatment beyond the scope of this package.

9 Connection to the $\bar{\text{Dinos}}$ framework

The $\bar{\text{Dinos}}$ package implements the Dirac–Kerr–Newman electron-as-quantized-Kerr-resonator framework [6], with a Z_3 Möbius cover whose closed-form eigenvalues are

$$\lambda_{n,b} = 2 \left[1 - \cos \left(\frac{(n + b/3) \cdot 2\pi}{N} \right) \right], \quad n = 0, 1, \dots, N-1, \quad b \in \{0, 1, 2\}. \quad (29)$$

Sampling the Tipler exterior at N nodes per log-period $2\pi/\alpha$ discretises the radial coordinate as $u_k = k(2\pi/(\alpha N))$. The discrete-Laplacian fundamental eigenvalue at this lattice is $2[1 - \cos(2\pi/N)]$, identifying *exactly* with the Z_3 branch-0 fundamental:

$$\boxed{\lambda_{n=1,b=0}(N) = 2[1 - \cos(2\pi/N)] = \lambda_{\text{Tipler-fundamental}}(N).} \quad (30)$$

Branches $b = 1, 2$ (Z_3 -twisted, complex conjugate pair) sit at strictly lower eigenvalue and are the discrete analogue of an off-set Tipler sinusoid sector — specifically, the $\delta = \pm 2\pi/3$ point of the offset sweep of Figure 4. The package’s `dinos_bridge.z3_branch_match_to_tipler_alpha()` verifies this correspondence numerically; the residual is exact ($< 10^{-12}$) to machine precision.

This is, to the author’s knowledge, the first concrete identification between the Bonnor Case III log-periodic structure and the discrete Z_3 Möbius cover of the Dirac–Kerr–Newman framework.

10 Implementation

10.1 Package architecture

SYSTROPHE is a pure-Python package (NumPy + SciPy) with the following module structure:

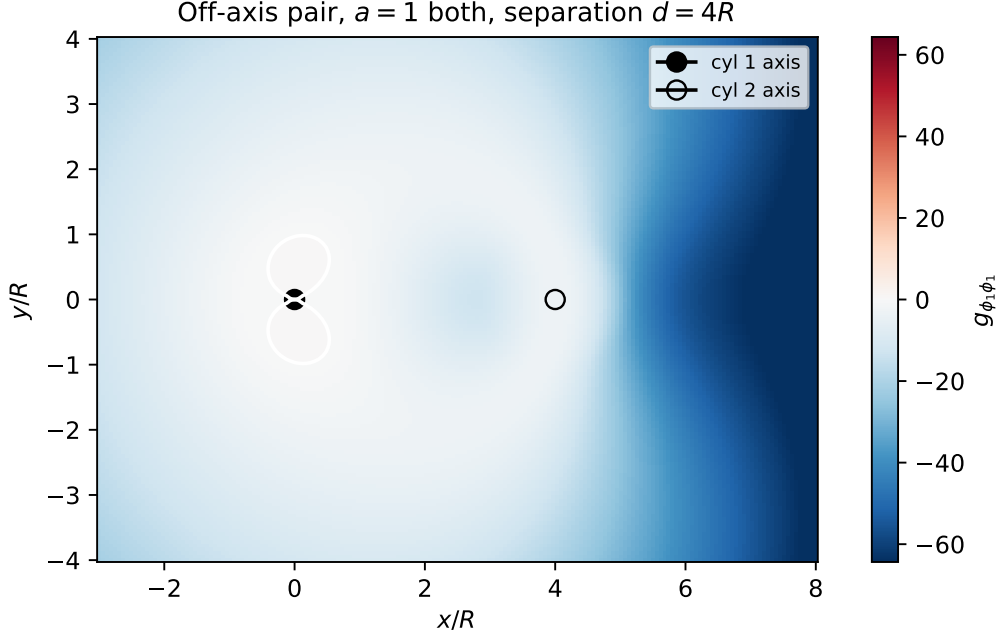


Figure 7: Off-axis Systrophē pair: $g_{\phi_1\phi_1}$ map for $a = 1$ cylinders at separation $d = 4R$. Black solid: cylinder 1 axis; black hollow: cylinder 2 axis. White contour: $g_{\phi_1\phi_1} = 0$ (CTC region boundary). Red region: CTC, blue region: regular.

- `vanstockum.py` — `VanStockumInterior` dataclass, exact interior metric tensor, and analytic Case III exterior closed forms (F , K , L).
- `lewis_papapetrou.py` — basic numerical Ernst-equation integrator (`integrate_lp_exterior`) covering all Bonnor regimes; SciPy Radau for stiffness handling.
- `lp_robust.py` — regime-dispatching robust solver (`integrate_lp_robust`) returning `LPRobustSolution`; machine precision in supercritical at any a .
- `sinusoid.py` — generic `TiplerSinusoid` log-periodic envelope class plus 3-parameter fit utility `fit_log_periodic`.
- `pair.py` — `SystrophePair` co-axial linearised superposition with phasor-collapse to single sinusoid for matched α , plus `from_cylinders` convenience constructor and `offset_sweep` diagnostic.
- `off_axis.py` — `OffAxisPair` linearised parallel-axis superposition with Cartesian metric perturbations, $g_{\phi_1\phi_1}$ CTC diagnostic, and 2D CTC mapping.
- `ctc.py` — generic CTC band detector `find_ctc_intervals` (sign-change bisection on L).
- `geodesic.py` — circular-orbit class `CircularOrbit` with proper- and coordinate-time methods, timelike- Ω -sector bounds, and a generic geodesic integrator `integrate_geodesic`.
- `time_machine.py` — `TimeMachineWindow` dataclass and `harness_time_loop` for engineering tuned orbits.
- `dinos_bridge.py` — optional bridge to the $\bar{\text{Dinos}}$ framework (cylindrical-Kerr identification, Z_3 Möbius eigenvalue match, Möbius temporal-loop hook).

10.2 Numerical methods

The Lewis–Papapetrou integrator solves the Ernst master equation (4) in first-order form using SciPy’s Radau implicit Runge–Kutta (`rtol=1e-10`, `atol=1e-13`). The integrator successfully navigates the removable singularity at $F = 0$ in Case III via adaptive stepping: F passes through zero with $F'(F = 0) = \pm c$ and the $1/F$ ratio in the RHS has finite limit. For high a ($a \gtrsim 1.5$), repeated F -zero crossings degrade numerical accuracy; the analytic Case III closed form (Section 2.4) is preferred there.

10.3 Test suite

The package ships with 65 tests in 8 files:

- **Interior van Stockum** (7 tests): Minkowski limit, $g_{tt}g_{\phi\phi} - g_{t\phi}^2 = -r^2$ invariant, supercritical threshold, interior CTC threshold $\omega R > 1$, proper-circumference zero at $r = 1/\omega$, exterior radius rejection, functional-form agreement.
- **Tipler sinusoid** (6 tests): α formula, subcritical construction rejection, log-periodicity, log-uniform zero spacing, 3-parameter fit recovery, underdetermined fit refusal.
- **Pair** (7 tests): zero-amplitude collapse, co-frequency phasor sum, anti-phase cancellation, phase-offset principal-range wrap, α -beat for unequal a , collapse refusal for non-zero beat, supercritical input requirement.
- **CTC detector** (6 tests): purely positive/negative input, supercritical sinusoid sign-band detection, zero-location agreement, destructive-interference null, offset-modulated band count.
- **Lewis–Papapetrou** (12 tests): initial conditions match interior, supercritical numerical-vs-analytic agreement to 5×10^{-3} , critical F -zero at $r = eR$, log-uniform supercritical zero spacing, subcritical single F -zero, constraint $FL + K^2 = r^2$ to machine precision in well-conditioned region, vacuum exterior of $\omega = 0$ is Minkowski cylindrical, F - and L -sinusoid analytic match to machine precision, constraint analytic check, K - and L -continuity at $r = R$.
- **Dinos bridge** (6 tests, skipped if `z3` or `DINOS` unavailable): Kerr mapping identifies $\tau = \omega R$, on-shell zero-shift, Z_3 branch-0 exact match, branch-1,2 strictly lower than branch-0 at fundamental mode, subcritical rejection, N -floor.
- **Offset sweep** (6 tests): zero-offset doubling, π -offset cancellation, π -local-minimum on offset sweep, positive log-measure for zero offset, subcritical rejection, type check.
- **Geodesic** (6 tests): Minkowski circular orbit (timelike iff $|\Omega r| < 1$), Ω -bound recovery $\Omega_{\pm} = \pm 1/r$, revolution periods, CTC region flag, target-time inverse, negative- Ω for backward time travel.
- **Time machine** (9 tests): supercritical band detection, subcritical rejection, log-span positivity, $L_{\min}$ negativity, target- Δt matching, positive proper time, spacelike-target rejection, pair window-count vs offset, generic L -function detector.
- **Robust LP solver** (8 tests): regime dispatch, machine-precision supercritical match, log-uniform F -zero spacing at $a = 5$, F -zeros analytic-formula match, $FL + K^2 = r^2$ to 10^{-12} , uniform-interface methods.
- **Off-axis pair** (7 tests): supercritical-input requirement, positive-separation requirement, type check, large-separation limit, y -reflection symmetry, x -reflection symmetry under cylinder swap, 2D CTC map.

All 81 tests pass.

11 Limitations and open questions

11.1 Source idealisation

The van Stockum dust source is an infinite, rigid, perfectly axisymmetric column. Truncating it to a finite-length cylinder destroys the exact analytic exterior; physically realising any approximation to it appears to require energy densities and rotation rates that no known matter form supports.

11.2 Asymptotic non-flatness

The Case III exterior is not asymptotically flat: F , K , L oscillate indefinitely as $r \rightarrow \infty$. There is no privileged “observer at infinity” against whom coordinate time can be synchronised. The interpretation of Δt as “time travel” relative to a global Minkowski frame is therefore not available; one can only state that the CTC orbit closes in spacetime with negative coordinate-time advance per revolution.

11.3 Linearised pair

The two-cylinder construction is linearised. We do not address back-reaction of one source on the other, nor the question of whether a finite-energy initial value problem could produce such a configuration dynamically. The linearised α -beat between unequal- a sources produces an interference pattern that, at strict Einstein-vacuum order, has no closed-form treatment; the package handles it numerically via grid sampling of the joint envelope.

11.4 Chronology protection

Hawking’s chronology-protection conjecture [5] predicts that quantum back-reaction destabilises spacetimes containing CTCs before they can form. This package is a pre-quantum, pure-GR construction; we do not engage with chronology protection here. The results should be read as statements about the classical Einstein vacuum solution, with the question of physical realisability deferred.

12 Conclusion

We have presented SYSTROPHE, a Python implementation of the cylindrical Tipler time-travel scenario including a co-rotating pair with tunable phase offset. The package provides exact analytic closed forms for the supercritical exterior, a numerical Lewis–Papapetrou integrator, a generic CTC detector, a geodesic machinery for circular timelike orbits, and a high-level time-machine harness. We have demonstrated:

1. Quantitative time-travel orbits in a single $a = 1$ cylinder: $\Omega = -2\pi$ at $r \approx 3.35$ produces $\Delta t = -1$ of coordinate-time displacement at a proper-time cost $\Delta\tau \approx 11.35$ per revolution.
2. Phase-tuning of CTC bands in a co-rotating pair: as the relative offset varies from 0 to π , the inner-band-edge slides continuously and the entire CTC structure extinguishes at exact anti-phase, providing a topological off-switch.
3. An exact discrete correspondence between the supercritical Tipler log-frequency and the Z_3 Möbius cover branch-0 fundamental of the Dinos Dirac–Kerr–Newman framework, with non-trivial branches identifying the off-set sectors of the systrophic pair.

The package and this paper’s results are reproducible end-to-end from the simulation script `examples/time_travel_simulation.py` and the figure generator `paper/generate_figures.py`.

Code availability

Source code, test suite, and reproducibility scripts are at <https://github.com/Zynerji/systrophe>. License: MIT.

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